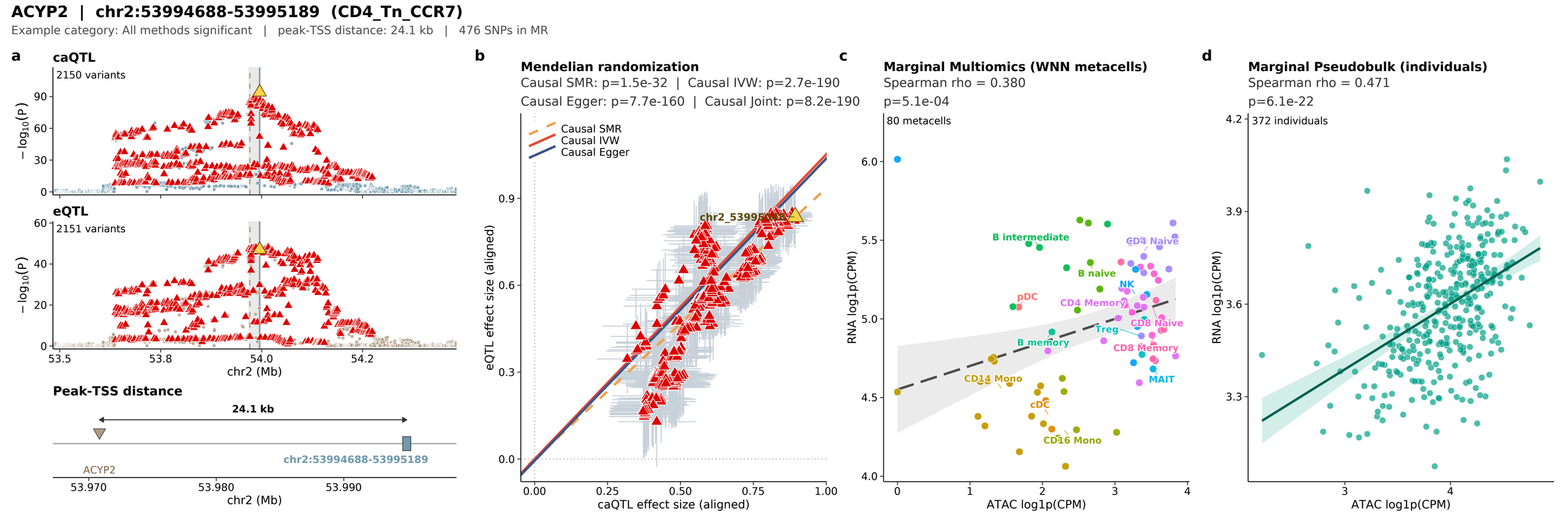
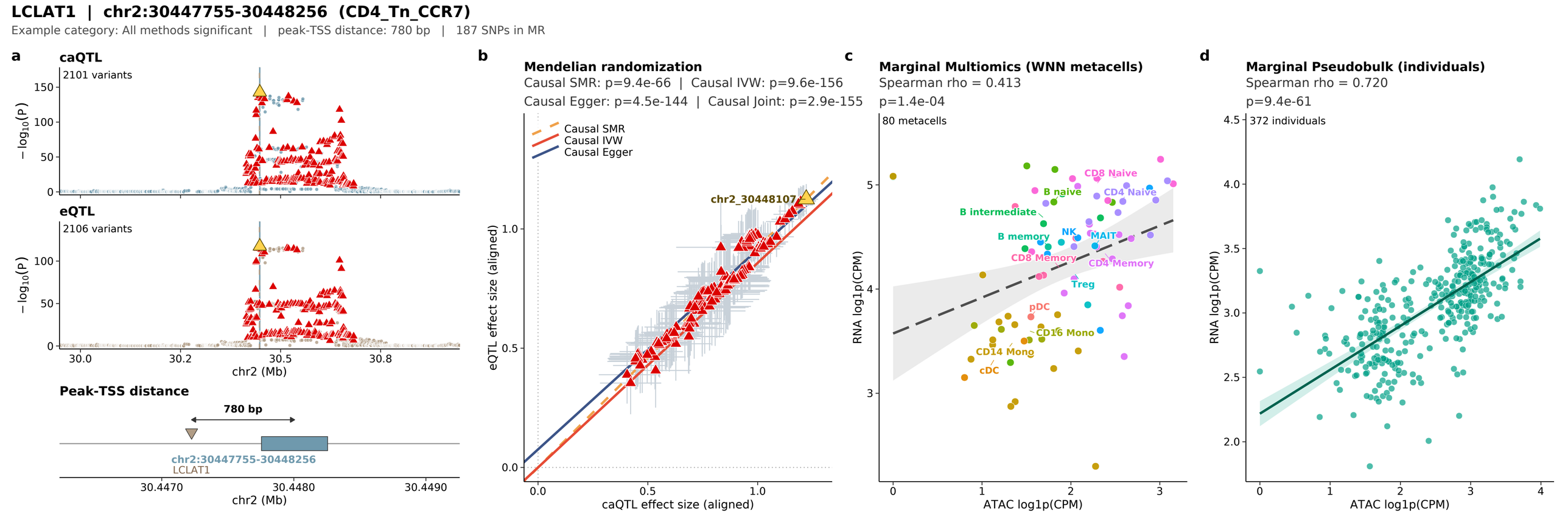
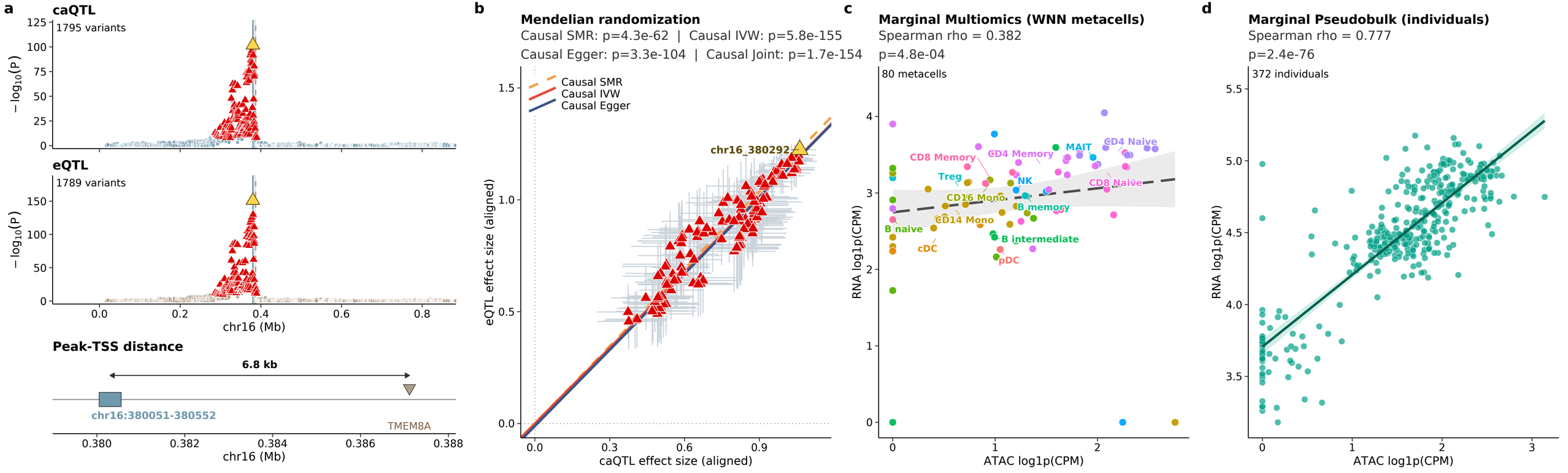

ACYP2 | chr2:53994688-53995189 (CD4_Tn_CCR7)
Example category: All methods significant | peak-TSS distance: 24.1 kb | 476 SNPs in MR
a
caQTL
2150 variants
eQTL
2151 variants
chr2 (Mb)
Peak-TSS distance
24.1 kb
chr2:53994688-53995189
ACYP2
b
Mendelian randomization
Causal SMR: p=1.5e-32 | Causal IVW: p=2.7e-190
Causal Egger: p=7.7e-160 | Causal Joint: p=8.2e-190
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.380
p=5.1e-04
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
B intermediate
CD4 Naive
B naive
NK
pDC
CD4 Memory
CD8 Naive
Treg
B memory
CD8 Memory
MAIT
CD14 Mono
cDC
CD16 Mono
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.471
p=6.1e-22
372 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


LCLAT1 | chr2:30447755-30448256 (CD4_Tn_CCR7)
Example category: All methods significant | peak-TSS distance: 780 bp | 187 SNPs in MR
a
caQTL
2101 variants
eQTL
2106 variants
− log10(P)
chr2 (Mb)
Peak-TSS distance
780 bp
chr2:30447755-30448256
LCLAT1
b
Mendelian randomization
Causal SMR: p=9.4e-66 | Causal IVW: p=9.6e-156
Causal Egger: p=4.5e-144 | Causal Joint: p=2.9e-155
Causal SMR
Causal IVW
Causal Egger
chr2_30448107
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.413
p=1.4e-04
80 metacells
CD8 Naive
B naive
CD4 Naive
B intermediate
B memory
NK
MAIT
CD8 Memory
CD4 Memory
Treg
pDC
CD16 Mono
CD14 Mono
cDC
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.720
p=9.4e-61
372 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

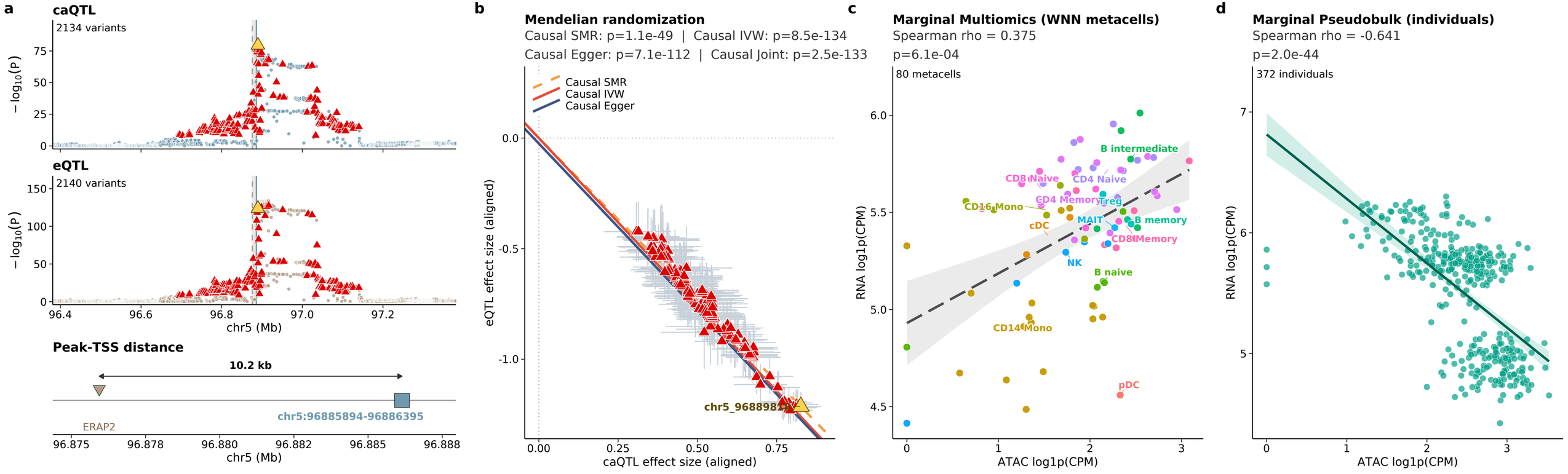
TMEM8A | chr16:380051-380552 (CD4_Tn_CCR7)

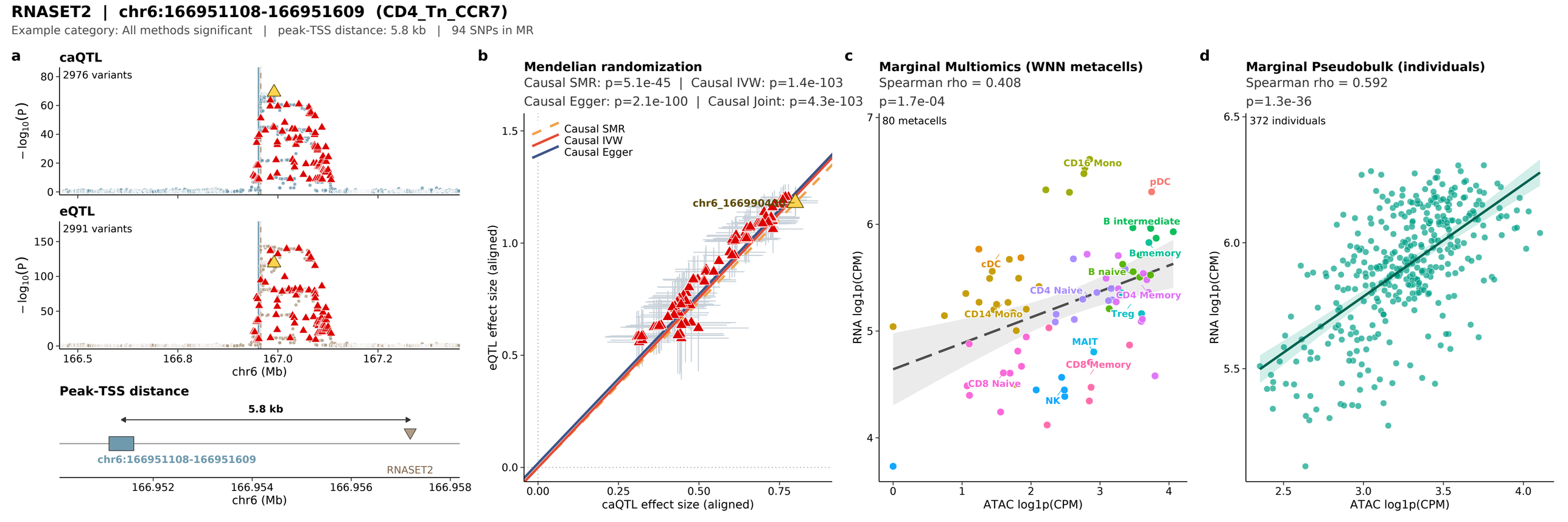
Example category: All methods significant | peak-TSS distance: 6.8 kb | 147 SNPs in MR

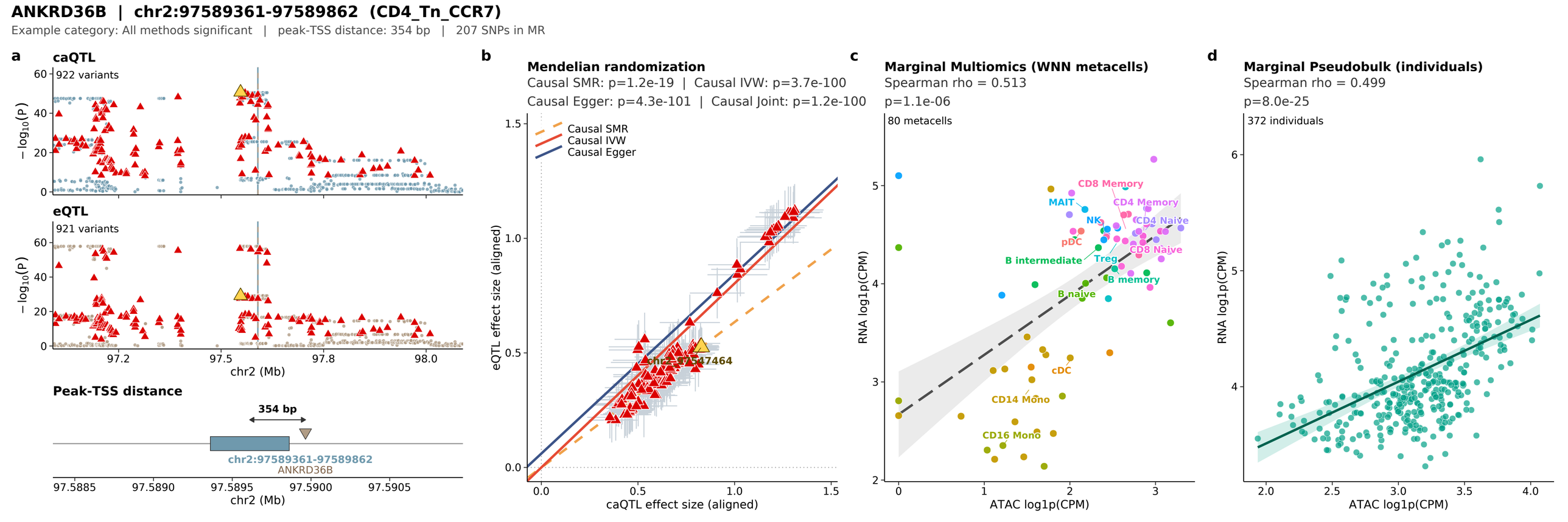


ERAP2 | chr5:96885894-96886395 (CD4_Tn_CCR7)

Example category: All methods significant / Opposite direction (MR vs. marginal) | peak-TSS distance: 10.2 kb | 148 SNPs in MR

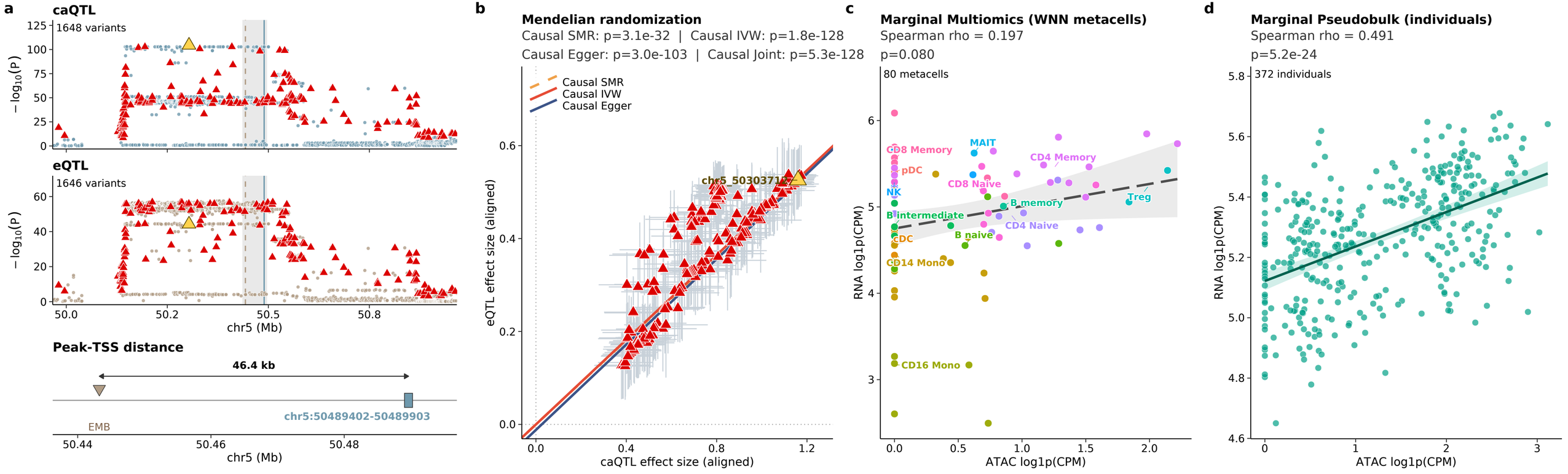



RNASET2 | chr6:166951108-166951609 (CD4_Tn_CCR7)
Example category: All methods significant | peak-TSS distance: 5.8 kb | 94 SNPs in MR
a
caQTL
2976 variants
− log10(P)
eQTL
2991 variants
chr6 (Mb)
Peak-TSS distance
5.8 kb
chr6:166951108-166951609
RNASET2
b
Mendelian randomization
Causal SMR: p=5.1e-45 | Causal IVW: p=1.4e-103
Causal Egger: p=2.1e-100 | Causal Joint: p=4.3e-103
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.408
p=1.7e-04
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
CD16 Mono
pDC
B intermediate
B memory
cDC
B naive
CD4 Naive
CD4 Memory
CD14 Mono
Treg
MAIT
CD8 Memory
CD8 Naive
NK
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.592
p=1.3e-36
372 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


ANKRD36B | chr2:97589361-97589862 (CD4_Tn_CCR7)
Example category: All methods significant | peak-TSS distance: 354 bp | 207 SNPs in MR
a
caQTL
922 variants
eQTL
921 variants
− log10(P)
chr2 (Mb)
Peak-TSS distance
354 bp
chr2:97589361-97589862
ANKRD36B
b
Mendelian randomization
Causal SMR: p=1.2e-19 | Causal IVW: p=3.7e-100
Causal Egger: p=4.3e-101 | Causal Joint: p=1.2e-100
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
chr2_97547464
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.513
p=1.1e-06
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
CD8 Memory
MAIT
CD4 Memory
NK
CD4 Naive
pDC
CD8 Naive
B intermediate
Treg
B memory
B naive
cDC
CD14 Mono
CD16 Mono
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.499
p=8.0e-25
372 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

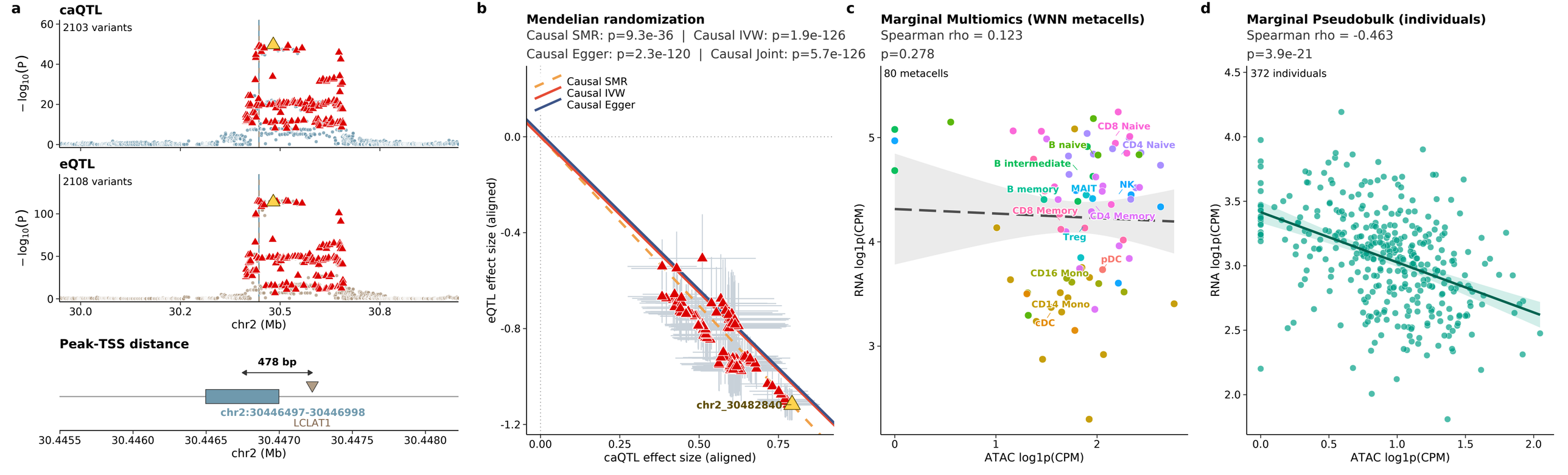
EMB | chr5:50489402-50489903 (CD4_Tn_CCR7)

Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 46.4 kb | 221 SNPs in MR



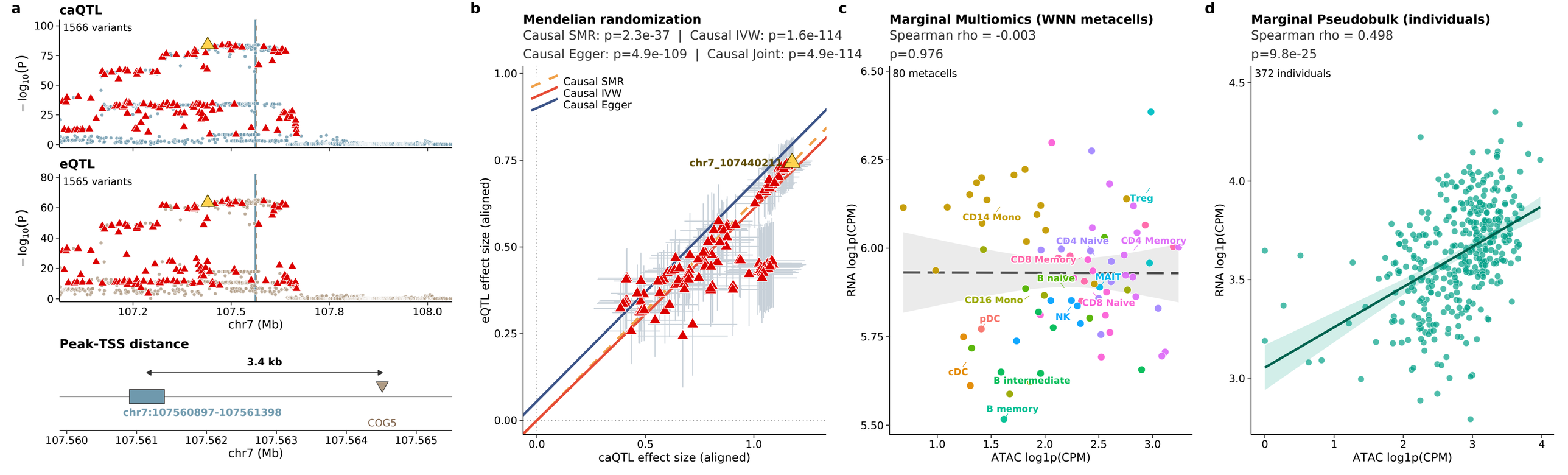
LCLAT1 | chr2:30446497-30446998 (CD4_Tn_CCR7)

Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 478 bp | 133 SNPs in MR



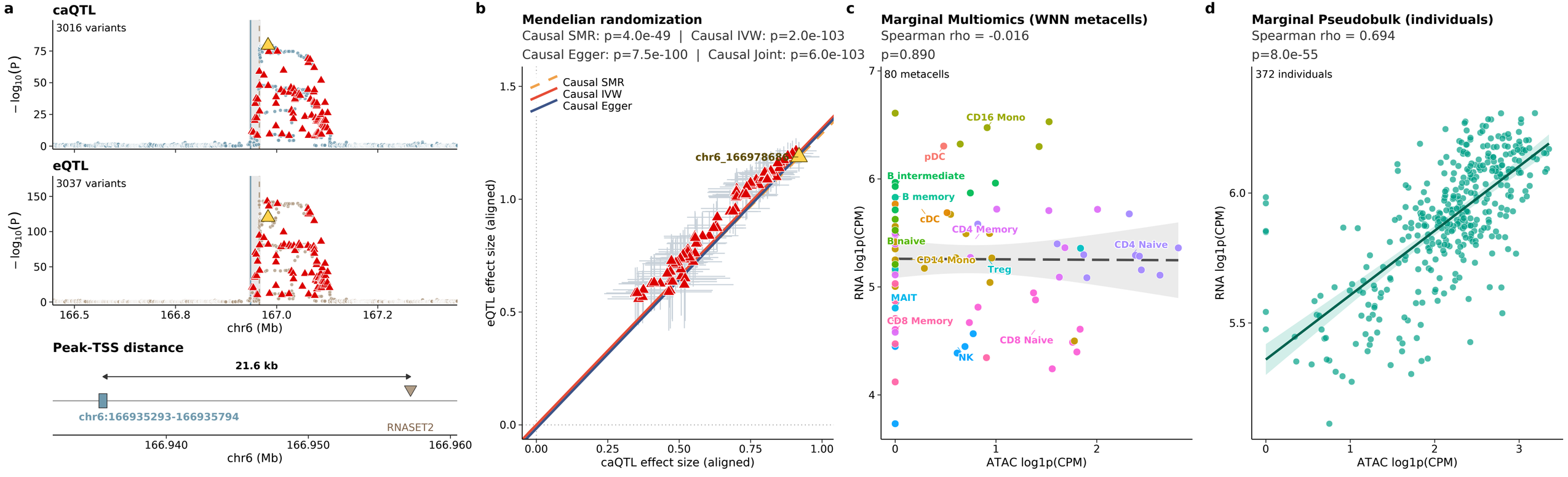
COG5 | chr7:107560897-107561398 (CD4_Tn_CCR7)

Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 3.4 kb | 140 SNPs in MR



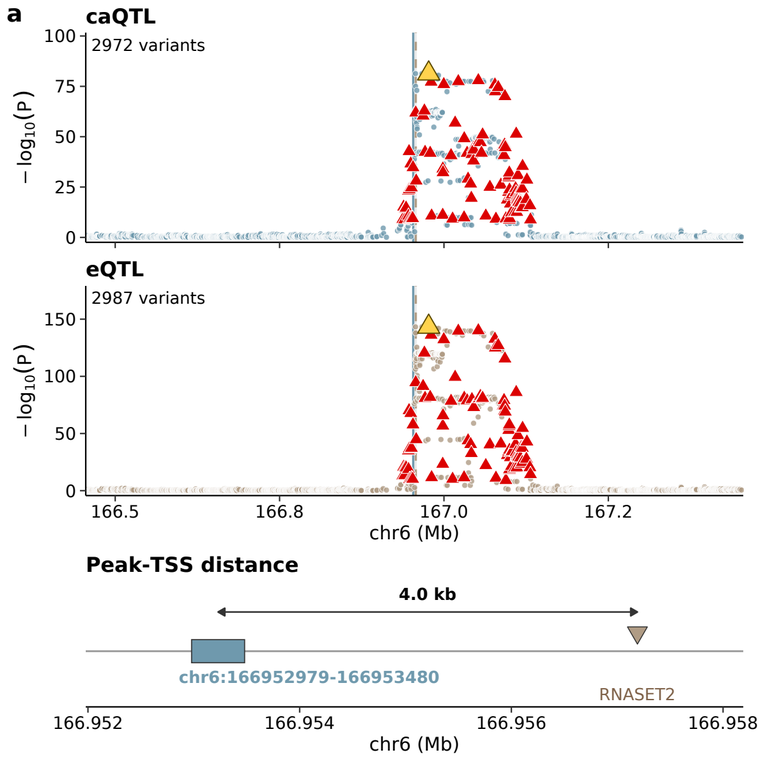
RNASET2 | chr6:166935293-166935794 (CD4_Tn_CCR7)

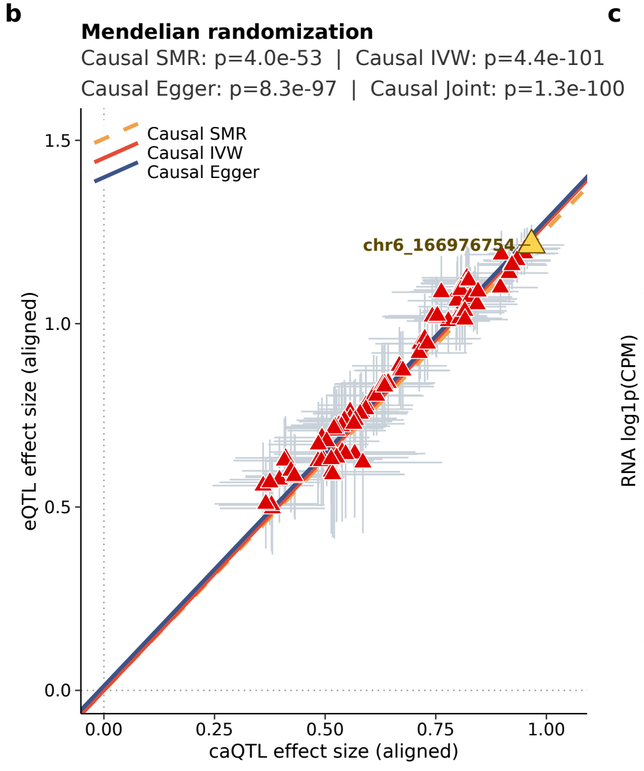
Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 21.6 kb | 92 SNPs in MR

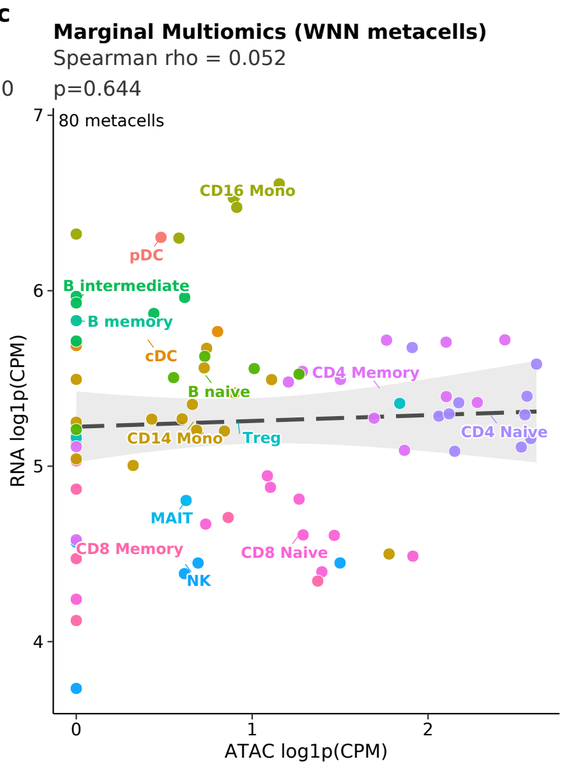


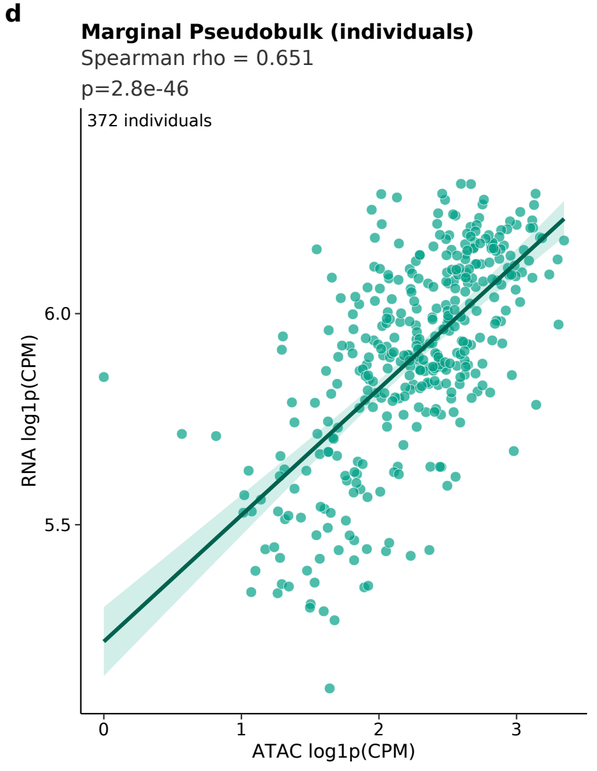
RNASET2 | chr6:166952979-166953480 (CD4_Tn_CCR7)

Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 4.0 kb | 95 SNPs in MR



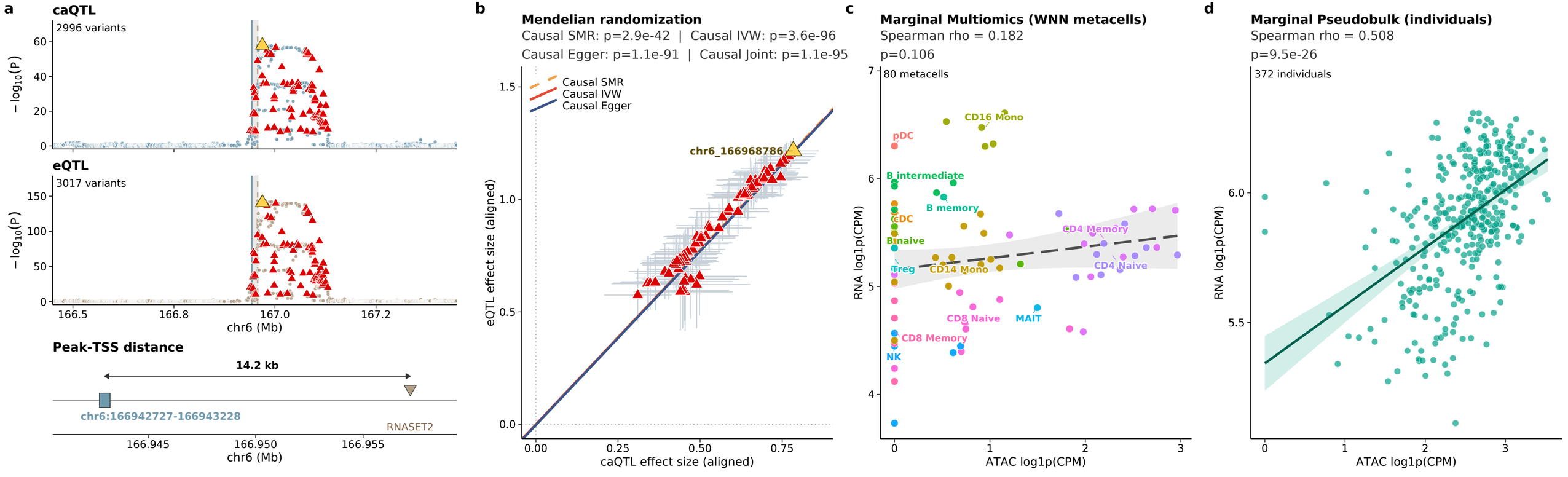


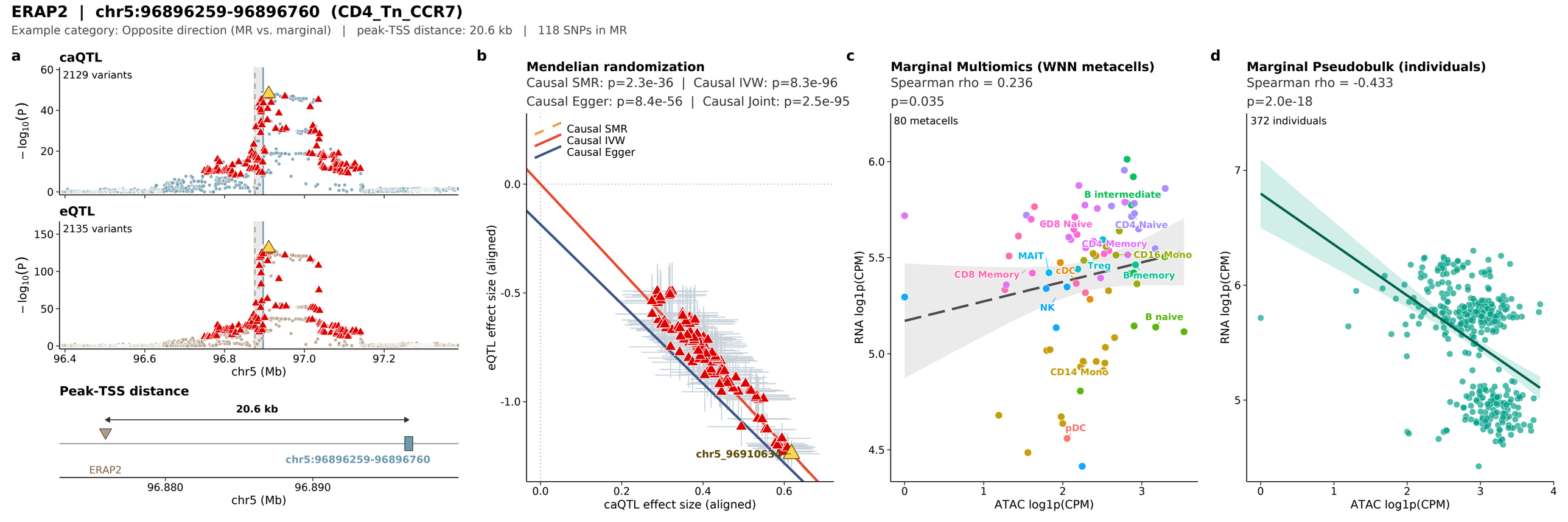


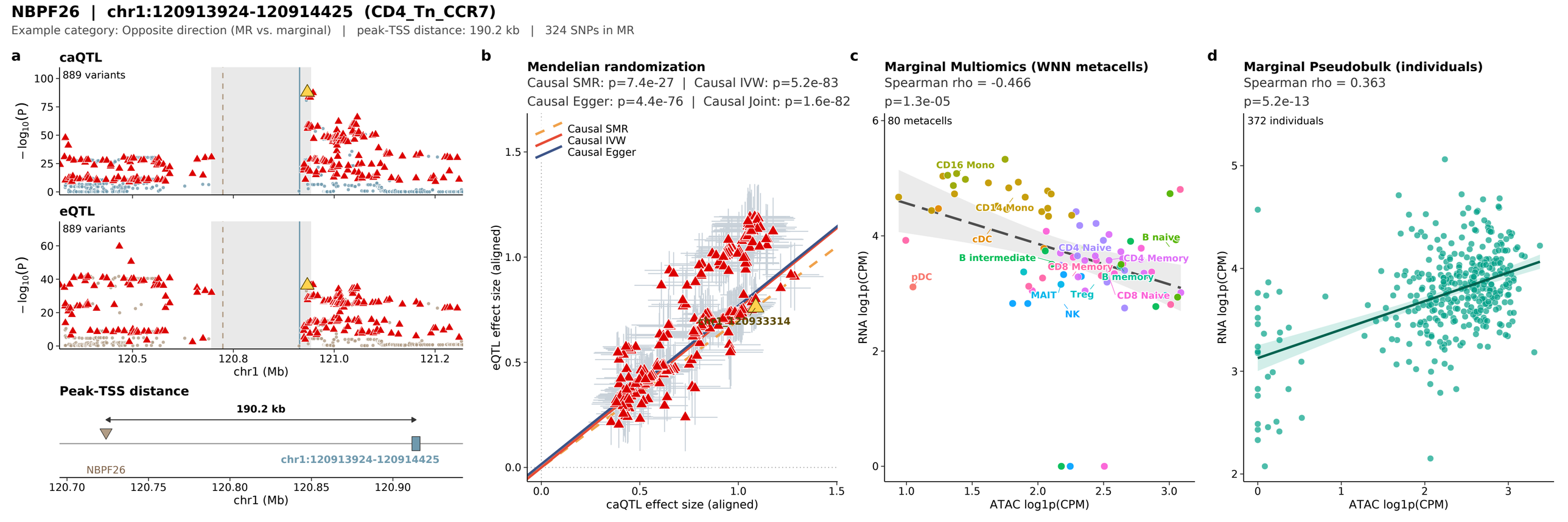


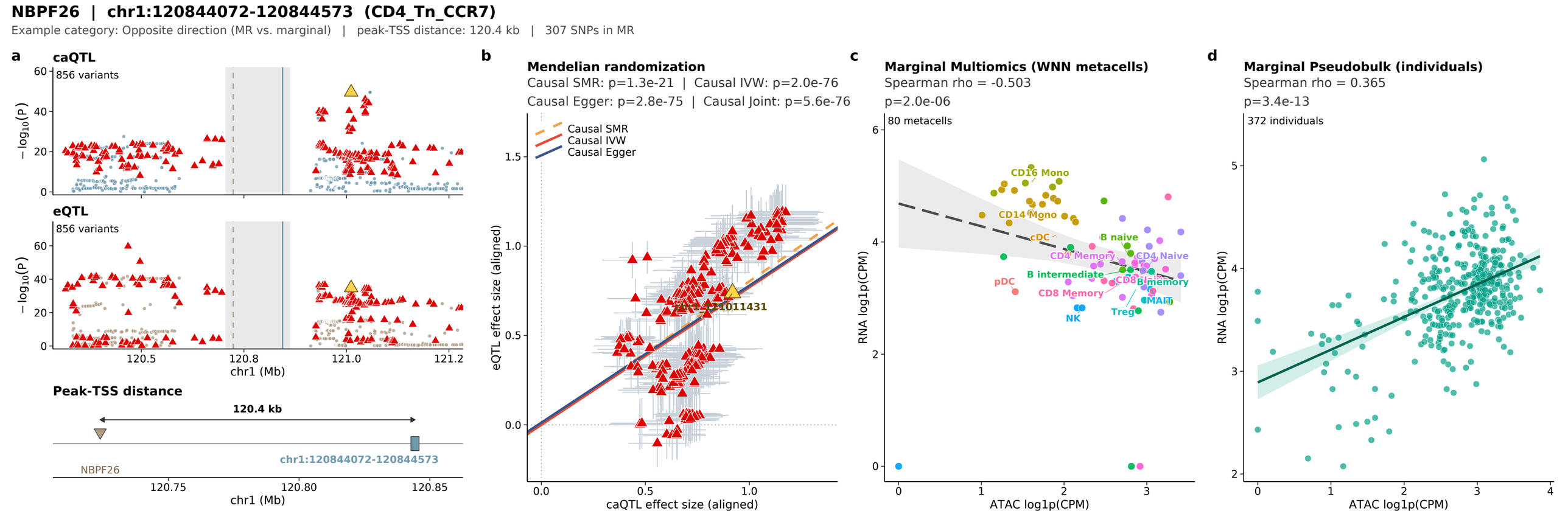
RNASET2 | chr6:166942727-166943228 (CD4_Tn_CCR7)

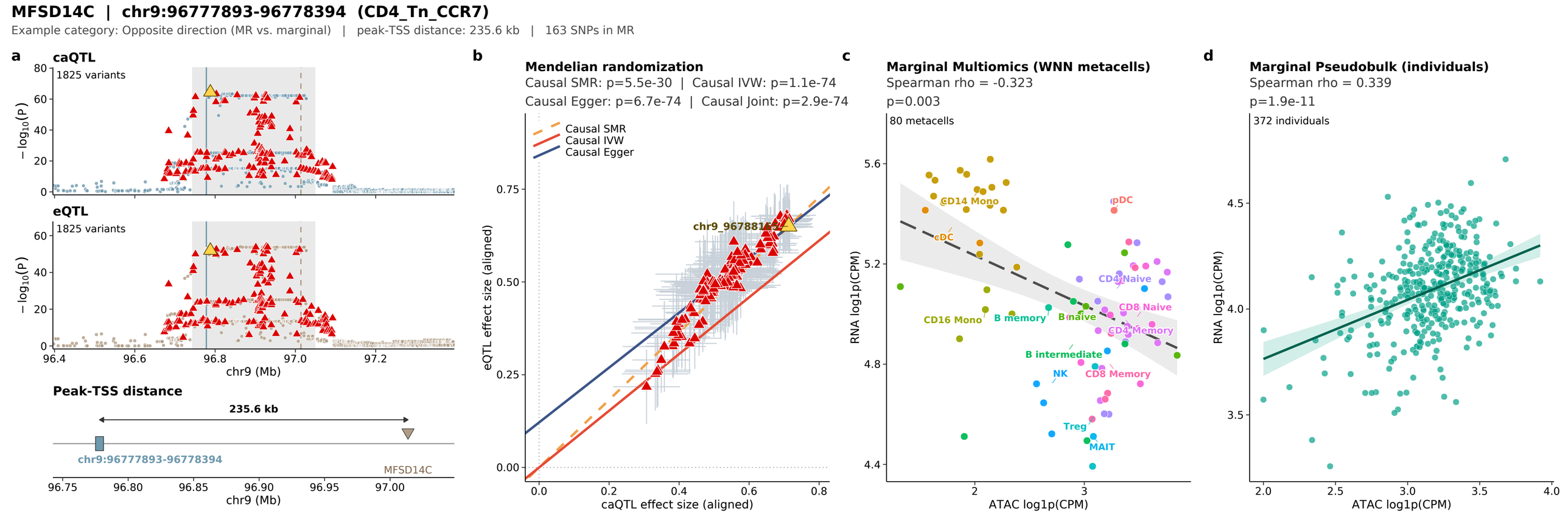
Example category: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 14.2 kb | 86 SNPs in MR

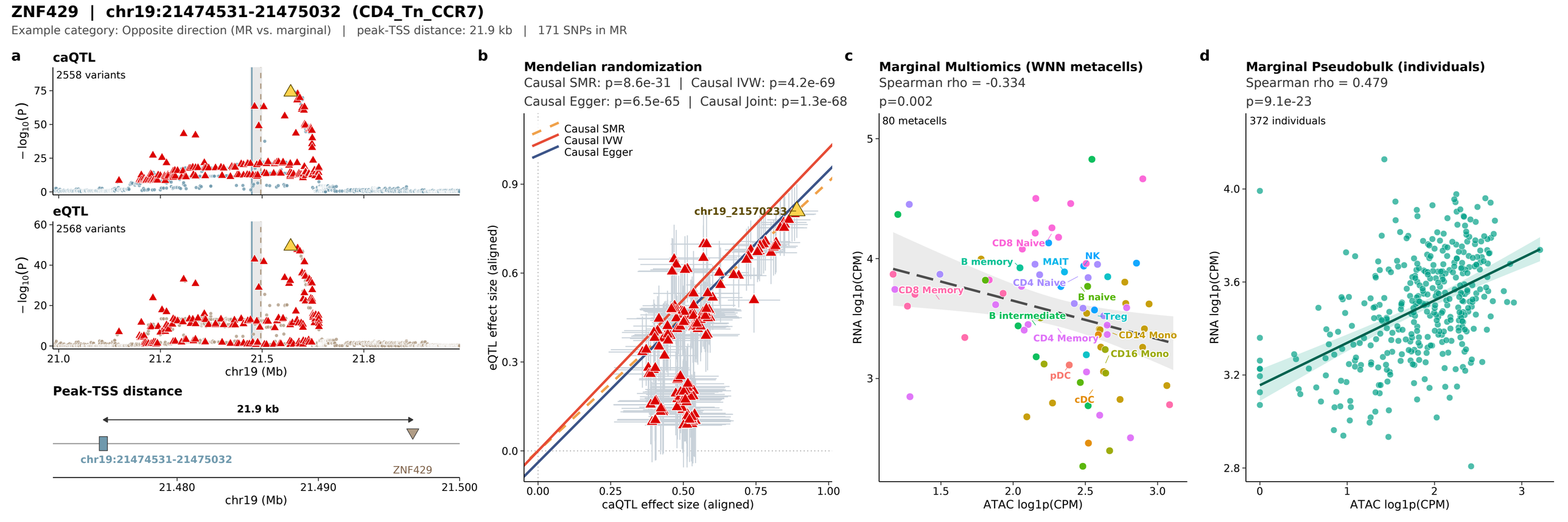


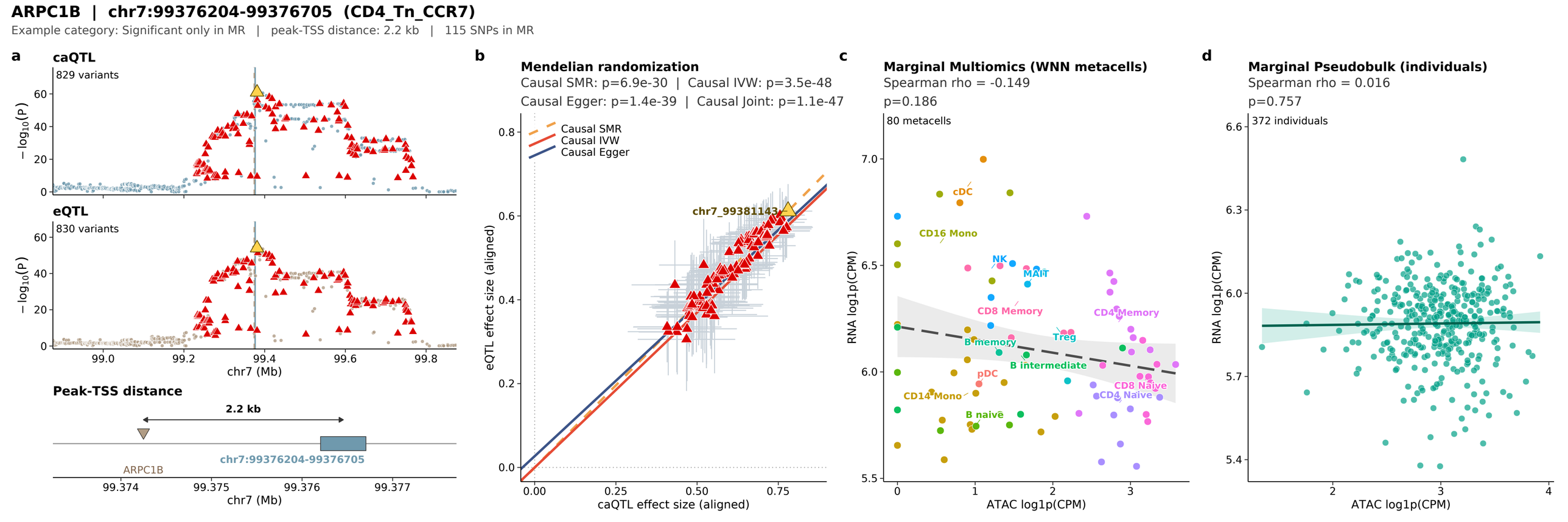

ERAP2 | chr5:96896259-96896760 (CD4_Tn_CCR7)
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 20.6 kb | 118 SNPs in MR
a
caQTL
2129 variants
eQTL
2135 variants
− log10(P)
chr5 (Mb)
Peak-TSS distance
20.6 kb
chr5:96896259-96896760
ERAP2
b
Mendelian randomization
Causal SMR: p=2.3e-36 | Causal IVW: p=8.3e-96
Causal Egger: p=8.4e-56 | Causal Joint: p=2.5e-95
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
chr5_96910634
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.236
p=0.035
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
B intermediate
CD8 Naive
CD4 Naive
CD4 Memory
CD16 Mono
MAIT
cDC
Treg
CD8 Memory
B memory
NK
B naive
CD14 Mono
pDC
d
Marginal Pseudobulk (individuals)
Spearman rho = -0.433
p=2.0e-18
372 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


NBPF26 | chr1:120913924-120914425 (CD4_Tn_CCR7)
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 190.2 kb | 324 SNPs in MR
a
caQTL
889 variants
eQTL
889 variants
− log10(P)
chr1 (Mb)
Peak-TSS distance
190.2 kb
NBPF26
chr1:120913924-120914425
b
Mendelian randomization
Causal SMR: p=7.4e-27 | Causal IVW: p=5.2e-83
Causal Egger: p=4.4e-76 | Causal Joint: p=1.6e-82
Causal SMR
Causal IVW
Causal Egger
chr1_120933314
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.466
p=1.3e-05
80 metacells
CD16 Mono
CD14 Mono
cDC
B intermediate
pDC
CD4 Naive
CD8 Memory
B naive
CD4 Memory
B memory
MAIT
Treg
CD8 Naive
NK
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.363
p=5.2e-13
372 individuals
RNA log1p(CPM)
ATAC log1p(CPM)


NBPF26 | chr1:120844072-120844573 (CD4_Tn_CCR7)
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 120.4 kb | 307 SNPs in MR
a
caQTL
856 variants
eQTL
856 variants
− log10(P)
chr1 (Mb)
Peak-TSS distance
120.4 kb
chr1:120844072-120844573
NBPF26
b
Mendelian randomization
Causal SMR: p=1.3e-21 | Causal IVW: p=2.0e-76
Causal Egger: p=2.8e-75 | Causal Joint: p=5.6e-76
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.503
p=2.0e-06
80 metacells
CD16 Mono
CD14 Mono
cDC
B naive
CD4 Memory
CD4 Naive
B intermediate
pDC
CD8 Memory
B memory
MAIT
NK
Treg
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.365
p=3.4e-13
372 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

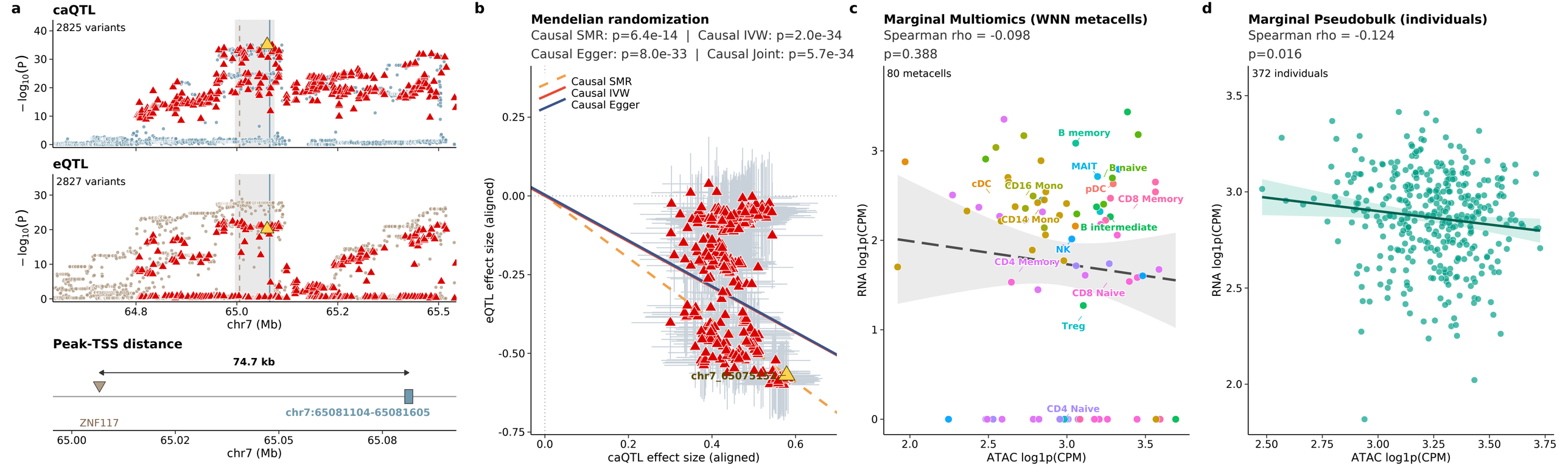

MFSD14C | chr9:96777893-96778394 (CD4_Tn_CCR7)
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 235.6 kb | 163 SNPs in MR
a
caQTL
1825 variants
eQTL
1825 variants
− log10(P)
chr9 (Mb)
Peak-TSS distance
235.6 kb
chr9:96777893-96778394
MFSD14C
b
Mendelian randomization
Causal SMR: p=5.5e-30 | Causal IVW: p=1.1e-74
Causal Egger: p=6.7e-74 | Causal Joint: p=2.9e-74
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.323
p=0.003
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
CD14 Mono
cDC
pDC
CD4 Naive
CD8 Naive
CD16 Mono
B memory
B naive
CD4 Memory
B intermediate
NK
CD8 Memory
Treg
MAIT
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.339
p=1.9e-11
372 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

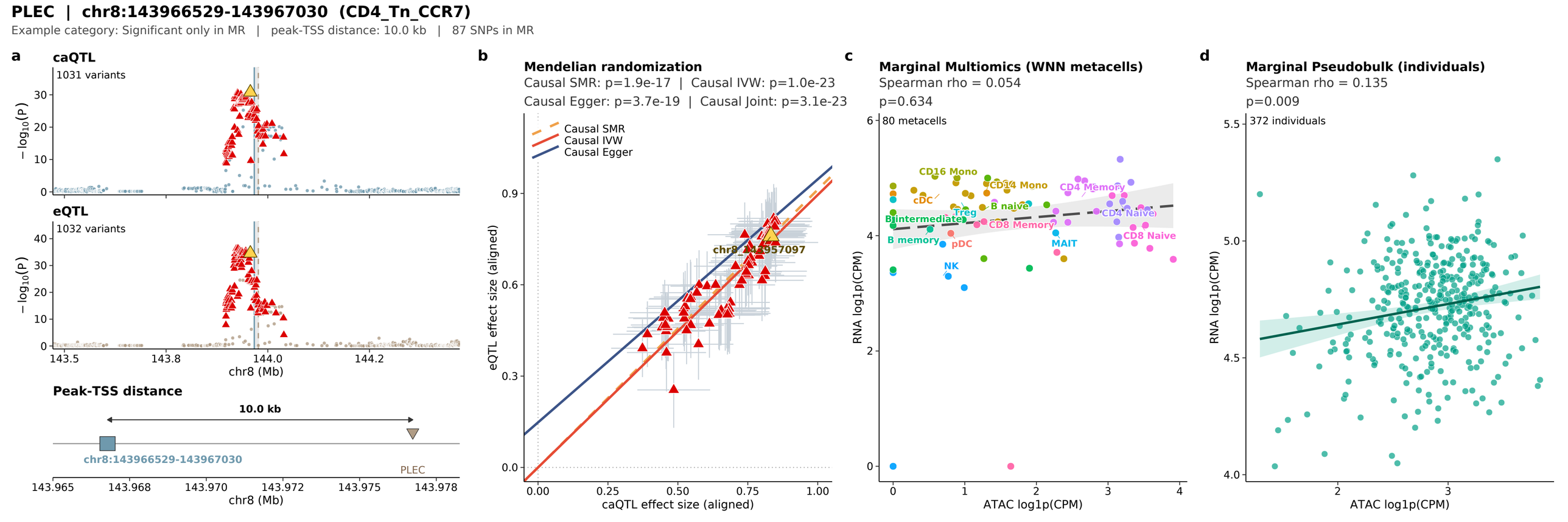

ZNF429 | chr19:21474531-21475032 (CD4_Tn_CCR7)
Example category: Opposite direction (MR vs. marginal) | peak-TSS distance: 21.9 kb | 171 SNPs in MR
a
caQTL
2558 variants
eQTL
2568 variants
− log10(P)
chr19 (Mb)
Peak-TSS distance
21.9 kb
chr19:21474531-21475032
ZNF429
b
Mendelian randomization
Causal SMR: p=8.6e-31 | Causal IVW: p=4.2e-69
Causal Egger: p=6.5e-65 | Causal Joint: p=1.3e-68
Causal SMR
Causal IVW
Causal Egger
chr19_21570233
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.334
p=0.002
80 metacells
CD8 Naive
B memory
MAIT
NK
CD4 Naive
CD8 Memory
B naive
B intermediate
Treg
CD4 Memory
CD14 Mono
CD16 Mono
pDC
cDC
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.479
p=9.1e-23
372 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

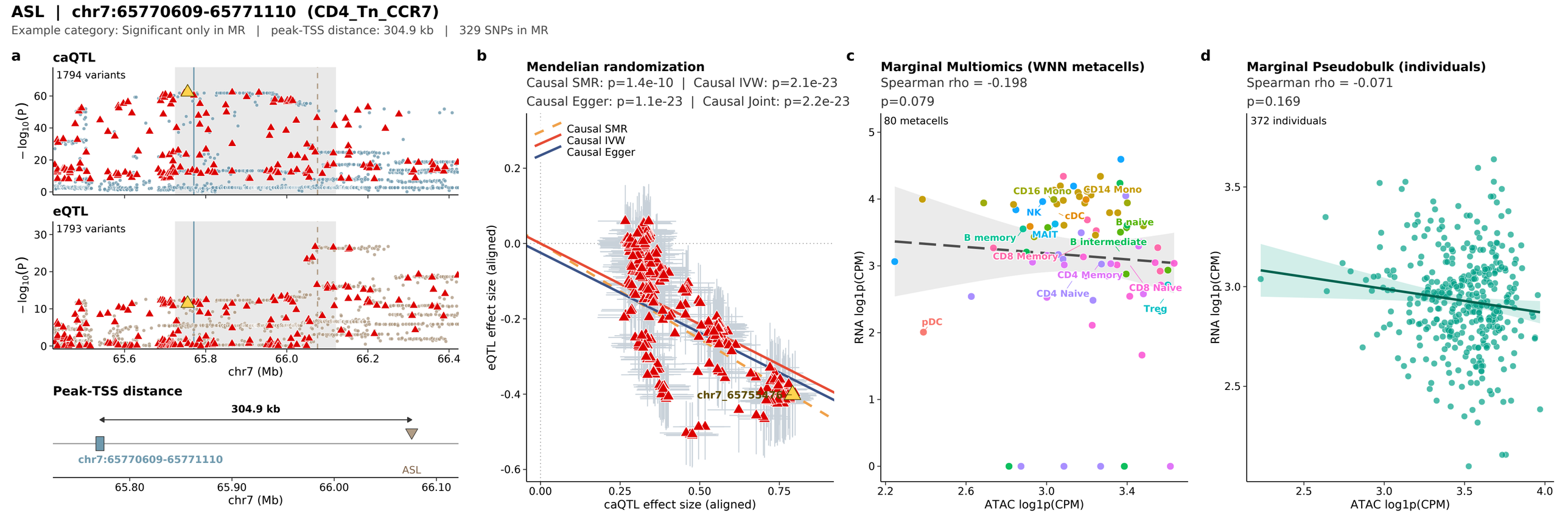

ARPC1B | chr7:99376204-99376705 (CD4_Tn_CCR7)
Example category: Significant only in MR | peak-TSS distance: 2.2 kb | 115 SNPs in MR
a
caQTL
829 variants
eQTL
830 variants
− log10(P)
chr7 (Mb)
Peak-TSS distance
2.2 kb
chr7:99376204-99376705
ARPC1B
b
Mendelian randomization
Causal SMR: p=6.9e-30 | Causal IVW: p=3.5e-48
Causal Egger: p=1.4e-39 | Causal Joint: p=1.1e-47
Causal SMR
Causal IVW
Causal Egger
chr7_99381143
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.149
p=0.186
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
cDC
CD16 Mono
NK
MAIT
CD8 Memory
CD4 Memory
Treg
B memory
B intermediate
pDC
CD8 Naive
CD14 Mono
CD4 Naive
B naive
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.016
p=0.757
372 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

ZNF117 | chr7:65081104-65081605 (CD4_Tn_CCR7)

Example category: Significant only in MR | peak-TSS distance: 74.7 kb | 392 SNPs in MR

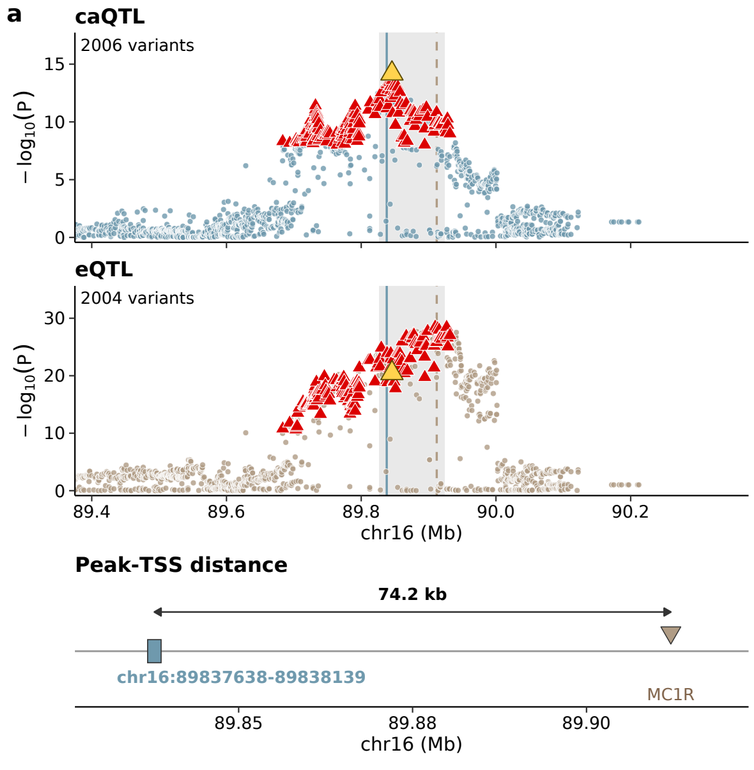


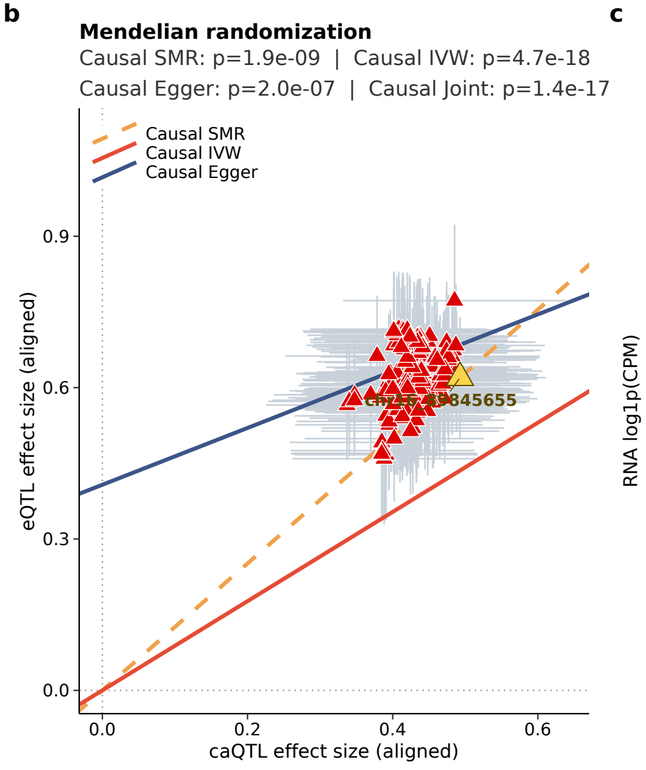

PLEC | chr8:143966529-143967030 (CD4_Tn_CCR7)
Example category: Significant only in MR | peak-TSS distance: 10.0 kb | 87 SNPs in MR
a
caQTL
1031 variants
eQTL
1032 variants
− log10(P)
chr8 (Mb)
Peak-TSS distance
10.0 kb
chr8:143966529-143967030
PLEC
b
Mendelian randomization
Causal SMR: p=1.9e-17 | Causal IVW: p=1.0e-23
Causal Egger: p=3.7e-19 | Causal Joint: p=3.1e-23
Causal SMR
Causal IVW
Causal Egger
chr8_144957097
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.054
p=0.634
80 metacells
CD16 Mono
CD14 Mono
CD4 Memory
cDC
Treg
B naive
CD4 Naive
Bintermediate
CD8 Memory
CD8 Naive
B memory
pDC
MAIT
NK
RNA log1p(CPM)
ATAC log1p(CPM)
d
Marginal Pseudobulk (individuals)
Spearman rho = 0.135
p=0.009
372 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

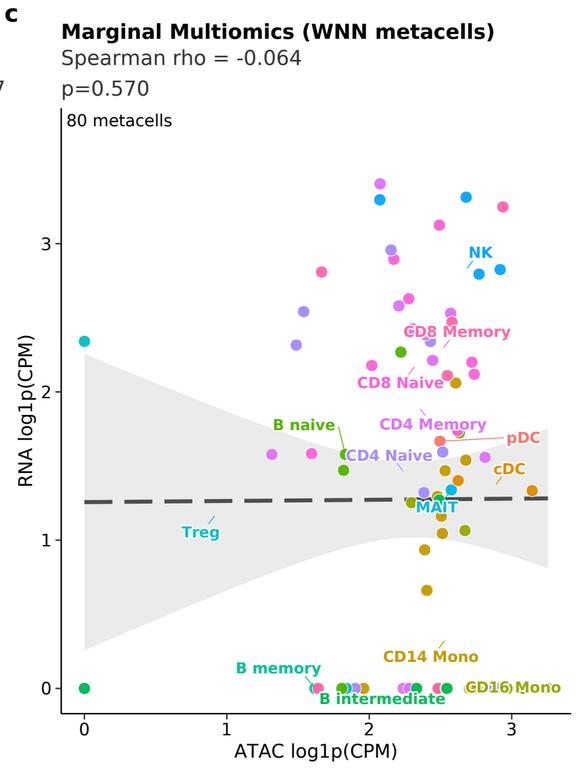

ASL | chr7:65770609-65771110 (CD4_Tn_CCR7)
Example category: Significant only in MR | peak-TSS distance: 304.9 kb | 329 SNPs in MR
a
caQTL
1794 variants
eQTL
1793 variants
− log10(P)
chr7 (Mb)
Peak-TSS distance
304.9 kb
chr7:65770609-65771110
ASL
b
Mendelian randomization
Causal SMR: p=1.4e-10 | Causal IVW: p=2.1e-23
Causal Egger: p=1.1e-23 | Causal Joint: p=2.2e-23
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
c
Marginal Multiomics (WNN metacells)
Spearman rho = -0.198
p=0.079
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
CD16 Mono
CD14 Mono
NK
cDC
B naive
B memory
MAIT
B intermediate
CD8 Memory
CD4 Memory
CD8 Naive
CD4 Naive
Treg
pDC
d
Marginal Pseudobulk (individuals)
Spearman rho = -0.071
p=0.169
372 individuals
RNA log1p(CPM)
ATAC log1p(CPM)

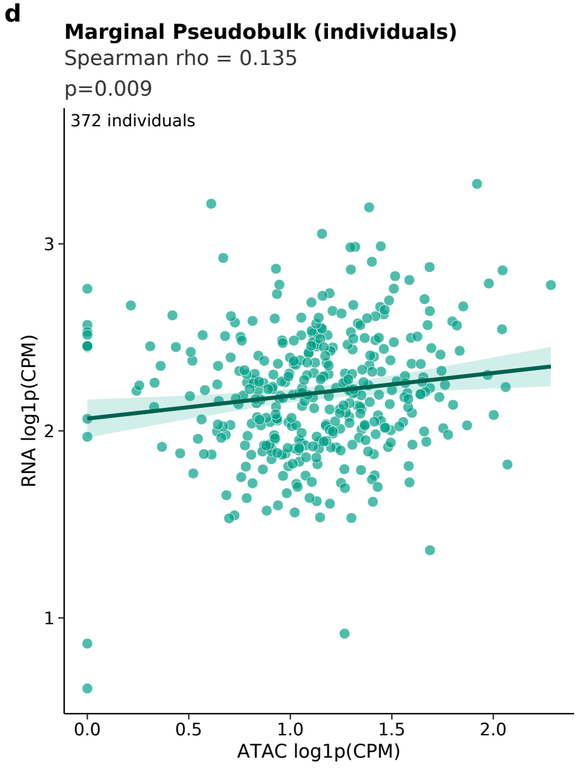
MC1R | chr16:89837638-89838139 (CD4_Tn_CCR7)

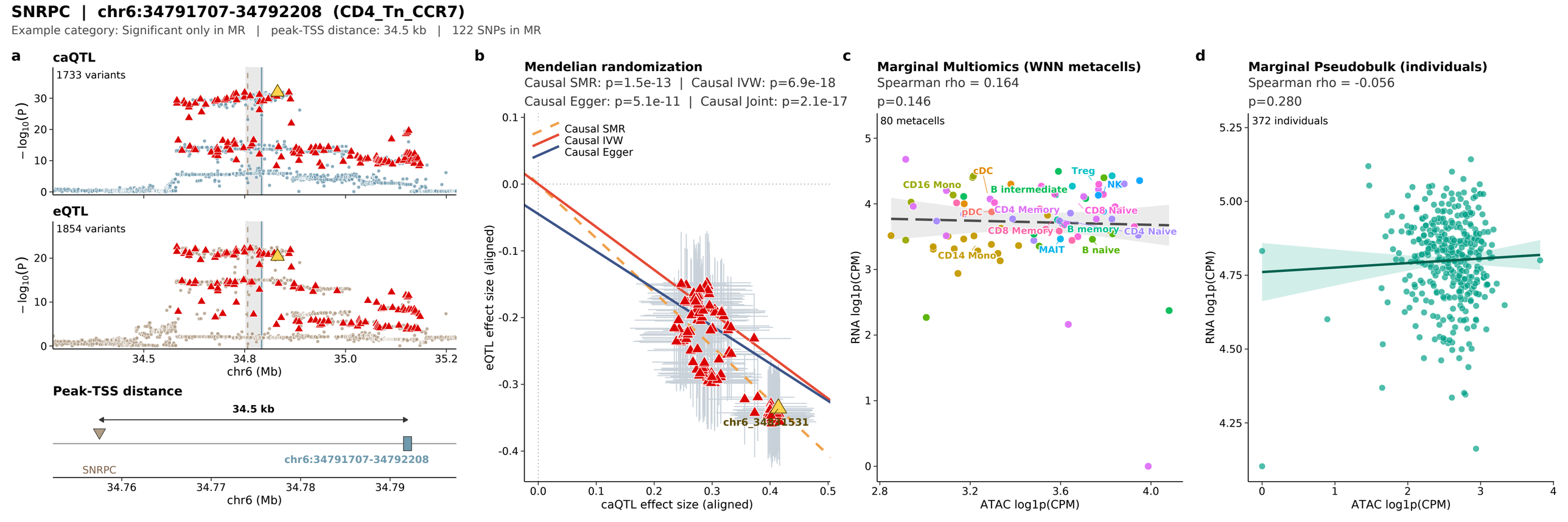
Example category: Significant only in MR | peak-TSS distance: 74.2 kb | 173 SNPs in MR










SNRPC | chr6:34791707-34792208 (CD4_Tn_CCR7)
Example category: Significant only in MR | peak-TSS distance: 34.5 kb | 122 SNPs in MR
a
caQTL
1733 variants
eQTL
1854 variants
− log10(P)
chr6 (Mb)
Peak-TSS distance
34.5 kb
chr6:34791707-34792208
SNRPC
b
Mendelian randomization
Causal SMR: p=1.5e-13 | Causal IVW: p=6.9e-18
Causal Egger: p=5.1e-11 | Causal Joint: p=2.1e-17
Causal SMR
Causal IVW
Causal Egger
eQTL effect size (aligned)
caQTL effect size (aligned)
chr6_34771531
c
Marginal Multiomics (WNN metacells)
Spearman rho = 0.164
p=0.146
80 metacells
RNA log1p(CPM)
ATAC log1p(CPM)
cDC
CD16 Mono
B intermediate
Treg
NK
pDC
CD4 Memory
CD8 Naive
CD8 Memory
B memory
CD4 Naive
MAIT
B naive
CD14 Mono
d
Marginal Pseudobulk (individuals)
Spearman rho = -0.056
p=0.280
372 individuals
RNA log1p(CPM)
ATAC log1p(CPM)